

THE AUTONOMOUS AGENT IN THE FINANCIAL MACHINE

POLICYARC: DETERMINISTIC AGENT SAFETY OS

Enterprise-grade spending limits, supervisor co-signatures, and zero-volatility micro-settlement on Circle's Arc L1.

THE ARC ADVANTAGE: On Circle's Arc L1, USDC is the native gas asset. Fees and settlement are strictly in US dollars.

PolicyArc pairs Arc's sub-second Malachite consensus (<800ms) with deterministic smart contract vaults to guarantee corporate treasury solvency, preventing prompt injection drains and runaway autonomous loops.

NATIVE GAS ASSET

100% Native USDC

Zero volatile tokens (ETH/SOL).
Predictable fractions of a cent per execution.

BFT CONSENSUS

< 800ms Finality

Malachite consensus engine allows
real-time agent micro-payments without lag.

SECURITY INVARIANTS

7 / 7 Verified

Audited under Build-Harness engineering
standard with 100% automated pass rate.

THE TWO FATAL CRISES OF AUTONOMOUS AGENT FINANCE

Why enterprises and developers cannot safely hand raw Web3 wallets to autonomous AI models:

CRISIS 01: GAS PRICE VOLATILITY LOCKUP

Fluctuating Network Fees Stalling Workflows

- **The Two-Token Hurdle:**

Agents on Ethereum/Solana must hold volatile native assets (ETH, SOL) solely to pay gas.

- **Sudden Fee Spikes:**

Network congestion spikes or depleted gas balances immediately stall autonomous task loops mid-run.

- **Unpredictable Accounting:**

Corporate finance teams cannot audit or budget AI agents when gas costs fluctuate 200% intra-day.

- **Idle Capital Friction:**

Treasuries must constantly top up gas balances across dozens of fragmented worker accounts.

CRISIS 02: UNBOUNDED TREASURY DRAINAGE

Unconstrained Signing Autonomy & Attacks

- **Binary Key Permissions:**

Standard private keys give agents 100% total signing authority with zero spend limits or gates.

- **Prompt Injection Hijacking:**

A malicious external API response can trick the agent into transferring all funds to an attacker.

- **Runaway Inference Loops:**

Code bugs or retry storms trigger recursive calls, draining thousands of dollars in minutes.

- **Permanent Irreversibility:**

Once funds leave a standard Web3 wallet, there is zero recourse or clawback mechanism.

WHY CIRCLE ARC L1 IS LOAD-BEARING FOR POLICYARC

Arc is not just another EVM layer; its unique primitives make sovereign agent economics mathematically viable:

PILLAR 01 // NATIVE GAS

Native USDC Gas

- Eliminates the two-token hurdle entirely.
- All transaction fees, gas accounting, and balances are denominated in US dollars.
- Agents only hold and budget a single, stable currency with zero exchange volatility.
- Gas costs are microscopic (~0.000120 USDC), enabling micro-payments down to \$0.05.

PILLAR 02 // SPEED

Malachite BFT (<800ms)

- AI agents operate in high-speed loops algorithmic requiring instant execution feedback.
- Arc provides deterministic finality in under 800ms via Malachite consensus.
- Zero multi-minute block confirmation lags or dropped replacement nonces.
- Enables real-time agent-to-agent and agent-to-API payment settlement pipelines.

PILLAR 03 // GLOBAL EXPANSION

Circle StableFX Ready

- Native on-chain atomic stablecoin the conversion engine built into settlement layer.
- Agents can hold USDC and seamlessly route payments in EURC or global stablecoins.
- Enables global compute leasing across European and Asian localized clusters.
- Future-proof substrate for enterprise multinational agent workforces.

THE THREE-TIER SAFETY VALVE & 7 INVARIANTS

PolicyVault implements deterministic on-chain governance backed by automated invariant testing:

ON-CHAIN GOVERNANCE ENGINE

Tier 1: Autonomous Spends ($\leq$ \$2.00 USDC)

Agent queries APIs or procures micro-tasks autonomously. Settles in $<800\text{ms}$ with native USDC receipt.

Tier 2: Supervisor Co-Signature Gate ($>$ \$2.00)

Preflight halts high-value spend. Requires an off-chain cryptographic ECDSA signature from human supervisor key.

Tier 3: Hard Ceilings (\$10.00 Cap / \$50 Daily)

Non-negotiable invariants. Any transaction above \$10 or daily spend above \$50 immediately reverts on-chain.

Emergency Circuit Breaker

Owner or supervisor can trigger fail-closed pause state instantly freezing all vault transfers.

BUILD-HARNESS: 7/7 INVARIANTS PASS (100%)

- INV-001: Per-Tx Spending Ceiling (\$10.00 cap) [PASS]
- INV-002: 24h Rolling Budget Accumulator (\$50.00) [PASS]
- INV-003: Supervisor ECDSA Dual-Signature Gateway [PASS]
- INV-004: Strict Destination Whitelist Registry [PASS]
- INV-005: Fail-Closed Emergency Circuit Breaker [PASS]
- INV-006: Reentrancy Protection via CEI Pattern [PASS]
- INV-007: Arc Native USDC Gas Predictability [PASS]

FRESH LIVE ARC TESTNET PROOFS:

- Tx 0xcc1e...f89b: $-\$0.25$ USDC Autonomous Micro-Spend
- Tx 0x0aae...78f9: $-\$3.50$ USDC Supervisor Co-Signed
- Contract: 0xb9176558920B53e9542AD75eDC9Bfa68de3A89C0
- Arcscan Verified Block #65,791,146 (Chain ID 5042002)

FROM MICROGRANT TO ENTERPRISE SCALE

The strategic roadmap positioning PolicyArc for the \$10,000–\$100,000 Circle Grant Program:

PHASE 01 (CURRENT)

Arc Microgrant MVP

- Complete working prototype on Arc Testnet.
- PolicyVault & MockUSDC contracts live.
- 7 Formally verified security invariants.
- Interactive operational console & docs.

PHASE 02 (Q4 2026)

Circle StableFX Integration

- Integrate Circle's native StableFX engine.
- Atomic multi-stablecoin swaps (USDC/EURC).
- Cross-border agent payroll settlement.
- Localized compute cluster leasing.

PHASE 03 (Q1 2027)

Departmental Pools & SDKs

- Hierarchical multi-agent corporate pools.
- Sub-agent micro-budget delegation.
- First-party plugins for LangChain & CrewAI.
- Audit & compliance export for corporate ERPs.

OFFICIAL SUBMISSION LINKS & VERIFICATION ARTIFACTS:

- Live Console: <https://policyarc.vercel.app>
- Interactive Docs: <https://policyarc.vercel.app/docs>
- GitHub Repository: <https://github.com/0xNexuz/policy-arc>
- PolicyVault: 0xb9176558920B53e9542AD75eDC9Bfa68de3A89C0